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Inventor(s)	JAFFE; Joel R.

GAMING SYSTEM AND METHOD WITH ENHANCED FREE-SPIN FEATURE BASED ON PERSISTENT ELEMENT INTERACTION

Abstract

A gaming machine and method for enhanced gameplay using persistent elements. Each persistent element is associated with a distinct accumulation condition and a respective game feature. The game logic animates accumulation and determines feature awards. Free spins are triggered by one persistent element. If other persistent elements are awarded simultaneously, the free spins are enhanced with features that modify symbol outcomes. The enhancements occur dynamically, based on the combination of awarded features, improving engagement and variation in gameplay.

Inventors: **JAFFE; Joel R. (Glenview, IL)**

Applicant: **LNW Gaming, Inc.** (Las Vegas, NV)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/169,242, filed on Feb. 15, 2023, titled “Gaming System and Method with a Persistent Element Feature,” which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0003] The present invention relates to a technological improvement to gaming systems, gaming machines, and methods and, more particularly, to technological improvements in connection with a persistent element feature.

BACKGROUND OF THE INVENTION

[0004] Gaming machines often include bonus features that enhance engagement and variability. Some games use persistent elements or “pots” that accumulate value or progress as gameplay continues. While common systems trigger features upon accumulation thresholds, they do not typically provide for coordination between such features or affect gameplay in nuanced ways, such as through conditional enhancements of free spins when features are awarded simultaneously. Improvements are desired that enhance player anticipation and interaction by modifying free spin behavior based on the state and timing of persistent elements.

SUMMARY OF THE INVENTION

[0005] A gaming machine and method are disclosed in which persistent elements accumulate in response to satisfaction of respective conditions during gameplay. Upon satisfaction of an accumulation condition, a persistent element may be animated and randomly awarded, triggering an associated game feature. A free-spin feature is initiated when a designated persistent element is awarded. When the free-spin feature is awarded simultaneously with one or more additional persistent elements, the free spins are enhanced based on the awarded features, modifying symbol outcomes during the free games.

[0006] Additional aspects of the invention will be apparent to those of ordinary skill in the art in view of the detailed description of various embodiments, which is made with reference to the drawings, a brief description of which is provided below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a front view of a free-standing gaming machine according to an embodiment of the present invention.

[0008] FIG. **2** is a block diagram of a gaming system according to an embodiment of the present invention.

[0009] FIGS. **3**A and **3**B are a flow diagram for a data processing method that corresponds to instructions executed by a controller, according to an embodiment of the present invention. FIG. **3**A generally relates to a base-game portion of a wagering game; FIG. **3**B generally relates to the awarding of one or more game features.

[0010] FIGS. **4-8** are exemplary presentations of game spin outcomes resulting from the flow diagram in FIGS. **3**A-**3**B.

[0011] FIGS. **9-12** are exemplary presentations of game spin outcomes in accordance with alternate embodiments of the invention.

[0012] While the invention is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail herein. It should be understood, however, that the invention is not intended to be limited to the particular forms disclosed. Rather, the invention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

[0013] While this invention is susceptible of embodiment in many different forms, there is shown in the drawings and will herein be described in detail preferred embodiments of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to the embodiments illustrated. For purposes of the present detailed description, the singular includes the plural and vice versa (unless specifically disclaimed); the words “and” and “or” shall be both conjunctive and disjunctive; the word “all” means “any and all”; the word “any” means “any and all”; and the word “including” means “including without limitation.”

[0014] For purposes of the present detailed description, the terms “wagering game,” “casino wagering game,” “gambling,” “slot game,” “casino game,” and the like include games in which a player places at risk a sum of money or other representation of value, whether or not redeemable for cash, on an event with an uncertain outcome, including without limitation those having some element of skill. In some embodiments, the wagering game involves wagers of real money, as found with typical land-based or online casino games. In other embodiments, the wagering game additionally, or alternatively, involves wagers of non-cash values, such as virtual currency, and therefore may be considered a social or casual game, such as would be typically available on a social networking web site, other web sites, across computer networks, or applications on mobile devices (e.g., phones, tablets, etc.). When provided in a social or casual game format, the wagering game may closely resemble a traditional casino game, or it may take another form that more closely resembles other types of social/casual games.

[0015] Referring to FIG. **1**, there is shown a gaming machine **10** similar to those operated in gaming establishments, such as casinos. With regard to the present invention, the gaming machine **10** may be any type of gaming terminal or machine and may have varying structures and methods of operation. For example, in some aspects, the gaming machine **10** is an electromechanical gaming terminal configured to play mechanical slots, whereas in other aspects, the gaming machine is an electronic gaming terminal configured to play a video casino game, such as slots, keno, poker, blackjack, roulette, craps, etc. The gaming machine **10** may take any suitable form, such as floor-standing models as shown, handheld mobile units, bartop models, workstation-type console models, etc. Further, the gaming machine **10** may be primarily dedicated for use in playing wagering games, or may include non-dedicated devices, such as mobile phones, personal digital assistants, personal computers, etc. Exemplary types of gaming machines are disclosed in U.S. Pat. Nos. 6,517,433, 8,057,303, and 8,226,459, which are incorporated herein by reference in their entireties.

[0016] The gaming machine **10** illustrated in FIG. **1** comprises a gaming cabinet **12** that securely houses various input devices, output devices, input/output devices, internal electronic/electromechanical components, and wiring. The cabinet **12** includes exterior walls, interior walls, and shelves for mounting the internal components and managing the wiring, and one or more front doors that are locked and require a physical or electronic key to gain access to the interior compartment of the cabinet **12** behind the locked door. The cabinet **12** forms an alcove **14** configured to store one or more beverages or personal items of a player. A notification mechanism **16**, such as a candle or tower light, is mounted to the top of the cabinet **12**. It flashes to alert an attendant that change is needed, a hand pay is requested, or there is a potential problem with the gaming machine **10**.

[0017] The input devices, output devices, and input/output devices are disposed on, and securely coupled to, the cabinet **12**. By way of example, the output devices include a primary presentation device **18**, a secondary presentation device **20**, and one or more audio speakers **22**. The primary presentation device **18** or the secondary presentation device **20** may be a mechanical-reel display device, a video display device, or a combination thereof. In one such combination disclosed in U.S. Pat. No. 6,517,433, a transmissive video display is disposed in front of the mechanical reel display to portray a video image superimposed upon electro-mechanical reels. In another combination disclosed in U.S. Pat. No. 7,654,899, a projector projects video images onto stationary or moving surfaces. In yet another combination disclosed in U.S. Pat. No. 7,452,276, miniature video displays are mounted to electro-mechanical reels and portray video symbols for the game. In a further combination disclosed in U.S. Pat. No. 8,591,330, flexible displays such as OLED or e-paper displays are affixed to electro-mechanical reels. The aforementioned U.S. Pat. Nos. 6,517,433, 7,654,899, 7,452,276, and 8,591,330 are incorporated herein by reference in their entireties.

[0018] The presentation devices **18**, **20**, the audio speakers **22**, lighting assemblies, and/or other devices associated with presentation are collectively referred to as a "presentation assembly" of the gaming machine **10**. The presentation assembly may include one presentation device (e.g., the primary presentation device **18**), some of the presentation devices of the gaming machine **10**, or all of the presentation devices of the gaming machine **10**. The presentation assembly may be configured to present a unified presentation sequence formed by visual, audio, tactile, and/or other suitable presentation means, or the devices of the presentation assembly may be configured to present respective presentation sequences or respective information.

[0019] The presentation assembly, and more particularly the primary presentation device **18** and/or the secondary presentation device **20**, variously presents information associated with wagering games, non-wagering games, community games, progressives, advertisements, services, premium entertainment, text messaging, emails, alerts, announcements, broadcast information, subscription information, etc. appropriate to the particular mode(s) of operation of the gaming machine **10**. The gaming machine **10** may include a touch screen(s) **24** mounted over the primary or secondary presentation devices, buttons **26** on a button panel, a bill/ticket acceptor **28**, a card reader/writer **30**, a ticket dispenser **32**, and player-accessible ports (e.g., audio output jack for headphones, video headset jack, USB port, wireless transmitter/receiver, etc.). It should be understood that numerous other peripheral devices and other elements exist and are readily utilizable in any number of combinations to create various forms of a gaming machine in accord with the present concepts.

[0020] The player input devices, such as the touch screen **24**, buttons **26**, a mouse, a joystick, a gesture-sensing device, a voice-recognition device, and a virtual-input device, accept player inputs and transform the player inputs to electronic data signals indicative of the player inputs, which correspond to an enabled feature for such inputs at a time of activation (e.g., pressing a "Max Bet" button or soft key to indicate a player's desire to place a maximum wager to play the wagering game). The inputs, once transformed into electronic data signals, are output to game-logic circuitry for processing. The electronic data signals are selected from a group consisting essentially of an electrical current, an electrical voltage, an electrical charge, an optical signal, an optical element, a magnetic signal, and a magnetic element.

[0021] The gaming machine **10** includes one or more value input/payment devices and value output/payout devices. In order to deposit cash or credits onto the gaming machine **10**, the value input devices are configured to detect a physical item associated with a monetary value that establishes a credit balance on a credit meter such as the "credits" meter **200** (see FIG. **4**). The physical item may, for example, be currency bills, coins, tickets, vouchers, coupons, cards, and/or computer-readable storage mediums. The deposited cash or credits are used to fund wagers placed on the wagering game played via the gaming machine **10**. Examples of value input devices include, but are not limited to, a coin acceptor, the bill/ticket acceptor **28**, the card reader/writer **30**, a wireless communication interface for reading cash or credit data from a nearby mobile device, and a network interface for withdrawing cash or credits from a remote account via an electronic funds transfer. In response to a cashout input that initiates a payout from the credit balance on the "credits" meter **200** (see FIG. **4**), the value output devices are used to dispense cash or credits from the gaming machine **10**. The credits may be exchanged for cash at, for example, a cashier or redemption station. Examples of value output devices include, but are not limited to, a coin hopper for dispensing coins or tokens, a bill dispenser, the card reader/writer **30**, the ticket dispenser **32** for printing tickets redeemable for cash or credits, a wireless communication interface for transmitting cash or credit data to a nearby mobile device, and a network interface for depositing cash or credits to a remote account via an electronic funds transfer.

[0022] Turning now to FIG. **2**, there is shown a block diagram of the gaming-machine architecture. The gaming machine **10** includes game-logic circuitry **40** securely housed within a locked box inside the gaming cabinet **12** (see FIG. **1**). The game-logic circuitry **40** includes a central processing unit (CPU) **42** connected to a main memory **44** that comprises one or more memory devices. The CPU **42** includes any suitable processor(s), such as those made by Intel and AMD. By way of example, the CPU **42** includes a plurality of microprocessors including a master processor, a slave processor, and a secondary or parallel processor. Game-logic circuitry **40**, as used herein, comprises any combination of hardware, software, or firmware disposed in or outside of the gaming machine **10** that is configured to communicate with or control the transfer of data between the gaming machine **10** and a bus, another computer, processor, device, service, or network. The game-logic circuitry **40**, and more specifically the CPU **42**, comprises one or more controllers or processors and such one or more controllers or processors need not be disposed proximal to one another and may be located in different devices or in different locations. The game-logic circuitry **40**, and more specifically the main memory **44**, comprises one or more memory devices which need not be disposed proximal to one another and may be located in different devices or in different locations. The game-logic circuitry **40** is operable to execute all of the various gaming methods and other processes disclosed herein. The main memory **44** includes a wagering-game unit **46**. In one embodiment, the wagering-game unit **46** causes wagering games to be presented, such as video poker, video blackjack, video slots, video lottery, etc., in whole or part.

[0023] The game-logic circuitry **40** is also connected to an input/output (I/O) bus **48**, which can include any suitable bus technologies, such as an AGTL+ frontside bus and a PCI backside bus. The I/O bus **48** is connected to various input devices **50**, output devices **52**, and input/output devices **54** such as those discussed above in connection with FIG. **1**. The I/O bus **48** is also connected to a storage unit **56** and an external-system interface **58**, which is connected to external system(s) **60** (e.g., wagering-game networks).

[0024] The external system **60** includes, in various aspects, a gaming network, other gaming machines or terminals, a gaming server, a remote controller, communications hardware, or a variety of other interfaced systems or components, in any combination. In yet other aspects, the external system **60** comprises a player's portable electronic device (e.g., cellular phone, electronic wallet, etc.) and the external-system interface **58** is configured to facilitate wireless communication and data transfer between the portable electronic device and the gaming machine **10**, such as by a near-field communication path operating via magnetic-field induction or a frequency-hopping spread spectrum RF signals (e.g., Bluetooth, etc.).

[0025] The gaming machine **10** optionally communicates with the external system **60** such that the gaming machine **10** operates as a thin, thick, or intermediate client. The game-logic circuitry **40**—whether located within ("thick client"), external to ("thin client"), or distributed both within and external to ("intermediate client") the gaming machine **10**—is utilized to provide a wagering game on the gaming machine **10**. In general, the main memory **44** stores programming for a random number generator (RNG), game-outcome logic, and game assets (e.g., art, sound, etc.)—all of which obtained regulatory approval from a gaming control board or commission and are verified by a trusted authentication program in the main memory **44** prior to game execution. The authentication program generates a live authentication code (e.g., digital signature or hash) from the memory contents and compare it to a trusted code stored in the main memory **44**. If the codes match, authentication is deemed a success and the game is permitted to execute. If, however, the codes do not match, authentication is deemed a failure that must be corrected prior to game execution. Without this predictable and repeatable authentication, the gaming machine **10**, external system **60**, or both are not allowed to perform or execute the RNG programming or game-outcome logic in a regulatory-approved manner and are therefore unacceptable for commercial use. In other words, through the use of the authentication program, the game-logic circuitry facilitates operation of the game in a way that a person making calculations or computations could not.

[0026] When a wagering-game instance is executed, the CPU **42** (comprising one or more processors or controllers) executes the RNG programming to generate one or more pseudo-random numbers. The pseudo-random numbers are divided into different ranges, and each range is associated with a respective game outcome. Accordingly, the pseudo-random numbers are utilized by the CPU **42** when executing the game-outcome logic to determine a resultant outcome for that instance of the wagering game. The resultant outcome is then presented to a player of the gaming machine **10** by accessing the associated game assets, required for the resultant outcome, from the main memory **44**. The CPU **42** causes the game assets to be presented to the player as outputs from the gaming machine **10** (e.g., audio and video presentations). Instead of a pseudo-RNG, the game outcome may be derived from random numbers generated by a physical RNG that measures some physical phenomenon that is expected to be random and then compensates for possible biases in the measurement process. Whether the RNG is a pseudo-RNG or physical RNG, the RNG uses a seeding process that relies upon an unpredictable factor (e.g., human interaction of turning a key) and cycles continuously in the background between games and during game play at a speed that cannot be timed by the player. Accordingly, the RNG cannot be carried out manually by a human and is integral to operating the game.

[0027] The gaming machine **10** may be used to play central determination games, such as electronic pull-tab and bingo games. In an electronic pull-tab game, the RNG is used to randomize the distribution of outcomes in a pool and/or to select which outcome is drawn from the pool of outcomes when the player requests to play the game. In an electronic bingo game, the RNG is used to randomly draw numbers that players match against numbers printed on their electronic bingo card.

[0028] The gaming machine **10** may include additional peripheral devices or more than one of each component shown in FIG. **2**. Any component of the gaming-machine architecture includes hardware, firmware, or tangible machine-readable storage media including instructions for performing the operations described herein. Machine-readable storage media includes any mechanism that stores information and provides the information in a form readable by a machine (e.g., gaming terminal, computer, etc.). For example, machine-readable storage media includes read only memory (ROM), random access memory (RAM), magnetic-disk storage media, optical storage media, flash memory, etc.

[0029] In accordance with various methods of conducting a wagering game on a gaming system in accord with the present concepts, the wagering game includes a game sequence in which a player makes a wager, and a wagering-game outcome is provided or displayed in response to the wager being received or detected. The wagering-game outcome, for that particular wagering-game instance, is then revealed to the player in due course following initiation of the wagering game. The method comprises the acts of conducting the wagering game using a gaming apparatus, such as the gaming machine **10** depicted in FIG. **1**, following receipt of an input from the player to initiate a wagering-game instance. The gaming machine **10** then communicates the wagering-game outcome to the player via one or more output devices (e.g., primary presentation device **18** or secondary presentation device **20**) through the presentation of information such as, but not limited to, text, graphics, static images, moving images, etc., or any combination thereof. In accord with the method of conducting the wagering game, the game-logic circuitry **40** transforms a physical player input, such as a player's pressing of a “Spin” touch key or button, into an electronic data signal indicative of an instruction relating to the wagering game (e.g., an electronic data signal bearing data on a wager amount).

[0030] In the aforementioned method, for each data signal, the game-logic circuitry **40** is configured to process the electronic data signal, to interpret the data signal (e.g., data signals corresponding to a wager input), and to cause further actions associated with the interpretation of the signal in accord with stored instructions relating to such further actions executed by the controller. As one example, the CPU **42** causes the recording of a digital representation of the wager in one or more storage media (e.g., storage unit **56**), the CPU **42**, in accord with associated stored instructions, causes the changing of a state of the storage media from a first state to a second state. This change in state is, for example, effected by changing a magnetization pattern on a magnetically coated surface of a magnetic storage media or changing a magnetic state of a ferromagnetic surface of a magneto-optical disc storage media, a change in state of transistors or capacitors in a volatile or a non-volatile semiconductor memory (e.g., DRAM, etc.). The noted second state of the data storage media comprises storage in the storage media of data representing the electronic data signal from the CPU **42** (e.g., the wager in the present example). As another example, the CPU **42** further, in accord with the execution of the stored instructions relating to the wagering game, causes the primary presentation device **18**, other presentation device, or other output device (e.g., speakers, lights, communication device, etc.) to change from a first state to at least a second state, wherein the second state of the primary presentation device comprises a visual representation of the physical player input (e.g., an acknowledgement to a player), information relating to the physical player input (e.g., an indication of the wager amount), a game sequence, an outcome of the game sequence, or any combination thereof, wherein the game sequence in accord with the present concepts comprises acts described herein. The aforementioned executing of the stored instructions relating to the wagering game is further conducted in accord with a random outcome (e.g., determined by the RNG) that is used by the game-logic circuitry **40** to determine the outcome of the wagering-game instance. In at least some aspects, the game-logic circuitry **40** is configured to determine an outcome of the wagering-game instance at least partially in response to the random parameter.

[0031] In one embodiment, the gaming machine **10** and, additionally or alternatively, the external system **60** (e.g., a gaming server), means gaming equipment that meets the hardware and software requirements for fairness, security, and predictability as established by at least one state's gaming control board or commission. Prior to commercial deployment, the gaming machine **10**, the external system **60**, or both and the casino wagering game played thereon may need to satisfy minimum technical standards and require regulatory approval from a gaming control board or commission (e.g., the Nevada Gaming Commission, Alderney Gambling Control Commission, National Indian Gaming Commission, etc.) charged with regulating casino and other types of gaming in a defined geographical area, such as a state. By way of non-limiting example, a gaming machine in Nevada means a device as set forth in NRS 463.0155, 463.0191, and all other relevant provisions of the Nevada Gaming Control Act, and the gaming machine cannot be deployed for play in Nevada unless it meets the minimum standards set forth in, for example, Technical Standards 1 and 2 and Regulations 5 and 14 issued pursuant to the Nevada Gaming Control Act. Additionally, the gaming machine and the casino wagering game must be approved by the commission pursuant to various provisions in Regulation 14. Comparable statutes, regulations, and technical standards exist in or are used in other gaming jurisdictions, including for example GLI Standard #11 of Gaming Laboratories International (which defines a gaming device in Section 1.5) and N.J.S.A 5:12-23, 5:12-45, and all other relevant provisions of the New Jersey Casino Control

Act. As can be seen from the description herein, the gaming machine **10** may be regulatorily approved and thus implemented with hardware and software architectures, circuitry, and other special features that differentiate it from general-purpose computers (e.g., desktop PCs, laptops, and tablets).

[0032] Referring now to FIGS. **3**A-**3**B, there is shown a flow diagram representing one data processing method corresponding to at least some instructions stored and executed by the game-logic circuitry **40** in FIG. **2** to perform operations according to an embodiment of the present invention. The data processing method is described below in connection with the exemplary presentations of different spin outcomes in FIGS. **4-8**.

Game Play Initiation

[0033] Referring to FIG. **3**A, the data processing method commences at step **300**. At step **302**, the game-logic circuitry controls one or more presentation devices (e.g., mechanical-reel display device, video display device, or a combination thereof) to present a plurality of symbol-bearing reels, an array of symbol positions, and a plurality of persistent elements. Although the method is described with respect to one presentation device, it is to be understood that the presentation described herein may be performed by a presentation assembly including more than one presentation device. The symbol positions of the array may be arranged in a variety of configurations, formats, or structures and may comprise a plurality of rows and columns. The rows of the array are oriented in a generally horizontal direction, and the columns of the array are oriented in a generally vertical direction. The symbol positions in each row of the array are horizontally aligned with each other, and the symbol positions in each column of the array are vertically aligned with each other. Alternatively, the symbol positions may be arranged in a honeycomb configuration with adjacent columns vertically offset from each other by one-half symbol position or adjacent rows horizontally offset from each other by one-half symbol position. The number of symbol positions in different rows and/or different columns may vary from each other. The reels may be associated with the respective columns of the array such that the reels spin vertically, and each reel populates a respective column. In another embodiment, the reels may be associated with the respective rows of the array such that the reels spin horizontally, and each reel populates a respective row. In some embodiments, the reels are associated with respective individual symbol positions of the array such that each reel animates in place and populates only its respective symbol position. The symbol array configuration may vary between the base game (a spin of the game purchased via an accepted wager and not subject to any bonus rules) and any free games or other bonus games utilizing the array.

[0034] Referring to FIGS. **4-8**, which illustrate examples of the display at the conclusion of various representative game spins, the symbol array **210** in the base game has a three-by-five rectangular configuration, and each symbol position is associated with a respective independent reel. The reels bear a plurality of symbols that may, for example, include various base game symbols **10**, J, Q, K, A and a WILD symbol that can substitute for any of the base game symbols. The plurality of symbols may also include accumulation symbols, described below. An example of landed base game symbols is shown in FIG. **4**. The symbol array need not have a three-by-five rectangular configuration. As one example, FIGS. **11-14** illustrate a four-by-five symbol array variation.

[0035] Each reel may contain one or more stacks (i.e., clumps) of symbols that appear adjacent to each other along the reel. Some or all of a stack may land in the array when its reel stops spinning. A stack of symbols may consist of two, three, four, or more like adjacent symbols. For example, in FIG. **4**, two K symbols is illustrated in the third column of the symbol array **210** and three K symbols in the fourth column of the symbol array **210** are examples of landed symbols belonging to stacks.

[0036] Different game features of various types may be associated with the persistent elements **212**, **214** and **216**. In some embodiments, a game feature simply awards a prize, for example a fixed or progressive jackpot amount. In some embodiments, a game feature may comprise any type

of bonus game. Non-limiting examples of bonus games include a certain number of free games (i.e., spins of the reels), a “pick' em” bonus game, a wheel-spinning game, etc. In still other embodiments, a bonus game may be played to determine payment of prize, for example, a fixed or progressive jackpot. In some embodiments, a game feature may comprise an enhancement to the game. An enhancement may include, without limitation, pay table modifiers such as multipliers, increased values on value-bearing symbols, modification to the reels to include improved symbols, such as wild symbols, modification of symbol weights or the removal of certain “blocking symbols”, additional rows or columns added to the array, additional free spins, replacement symbols for symbols already present in the array, etc. The enhancement may be applied to the current base game outcome or may be applied to one or more subsequent bonus games or wagered base game plays, as will be shown. In some embodiments, a game feature may provide a second plurality of symbol-bearing reels and a second array of symbol positions to be filled, as described above, for a certain number of game cycles. Each game feature has a different impact on the expected value (EV) of the game. The relative frequency of the game features may be controlled by adjusting which coin pot they are associated with in the cascade sequence, for example. Thus, a less lucrative game feature may be won more frequently, for example, approximately once in every ten game spin cycles, while a higher paying game feature may only occur approximately once in every one hundred game spin cycles according to a weighted cascade value determination. This allows game features to be awarded more often, for player enjoyment, while maintaining the overall expected EV of the wagering game. In the examples described and illustrated by FIGS. **4-8** and the description of method **300**, a “Respin” game feature is associated with the first persistent element **212**, a “Wild Stack” game feature is associated with the second persistent element **214**, and a “Free Spins” game feature is associated with the third persistent element **216**.
[0037] In FIGS. **4-8**, the persistent elements are represented as triangles. The persistent elements **212**, **214** and **216** may take forms other than triangles, including, for example, unicorns, castles, coin pots, urns, vases, jars, jugs, cans, bowls, piggy banks, beehives, inflating balloons, ladders, dials, meters, etc. Similarly, the accumulation symbols associated with the persistent elements may take forms other than coins, including, for example, balloons, colored dollar signs, etc. The accumulation symbols may but need not be color-matched to their respective persistent elements, provided that their association to a persistent element is indicated. For example, the BLUE coins and the first persistent element **212** may be labeled or colored blue, the GOLD coins and the second persistent element **214** may be labeled or colored gold, and the RED coins and the third persistent element **216** may be labeled or colored red. Each accumulation symbol may have a unique form. In accordance with one or more embodiments, the accumulation symbol for each of the persistent elements may vary in form from one accumulation symbol to another. For example, one accumulation symbol may be a balloon while a second accumulation symbol may be a dollar sign and a third accumulation symbol may be coin, two accumulation symbols may be coins while a third accumulation symbol may be a dollar sign, etc.
[0038] At step **304**, the game-logic circuitry detects, via a value input device, a physical item associated with a monetary value that establishes a monetary balance in the form of cash or credits. In FIGS. **4-8**, the monetary balance may be shown on a meter **200**.
Game Spin
[0039] At step **306**, the game-logic circuitry initiates a game of a wagering game cycle (i.e., spin cycle) in response to an input indicative of a wager covered by the monetary balance. To initiate a spin of the reels, the player may press a “Spin” or “Max Bet” key on a button panel or touch screen. In FIGS. **4-8** the wager may be shown on a bet meter **202**.
[0040] At step **308**, using an RNG, the game-logic circuitry spins and stops the reels to randomly land symbols from the reels in the array in visual association with one or more paylines (also known as lines, ways, patterns, or arrangements). The reel spin may be animated on a video display by depicting symbol-bearing strips moving vertically across the display and synchronously

updating the symbols visible on each strip as the strip moves across the display. Alternatively, the reels may be physical/electromechanical reels. FIGS. **4-8** depict various game spin outcomes.

[0041] At step **310**, the game-logic circuitry determines whether or not an accumulation condition appears in the array **210**. An accumulation condition may be an individual accumulation symbol, as in the coin examples described here or a designated symbol combination (described below). FIG. **5** illustrates that the plurality of symbols also includes accumulation symbols in the form of BLUE, GOLD and RED coin symbols. Players hope to land the BLUE, GOLD or RED coin symbols in the symbol array **210** to win the application of the game features associated with three persistent elements **212**, **214**, **216**. In the example of FIG. **5**, a BLUE COIN is associated with the first persistent element **212**, a GOLD coin is associated with the second persistent element **214**, and a RED coin is associated with the third persistent element **216**.

[0042] If no accumulation condition appears in the array **210**, the game-logic circuitry proceeds to step **312**, to be described later. If the game-logic circuitry determines that an accumulation condition appears in array **210**, flow proceeds to step **324** of FIG. **3**B, which will be described first.

Animation of Persistent Element Growth

[0043] During a player's gaming session, any growth in size of the persistent elements **212**, **214** and **216** persists from one wagering game cycle to the next such that the player perceives that a game feature corresponding to a persistent element may be getting closer to being awarded. When the size of the persistent elements has no bearing on whether the associated game feature will actually be awarded, this is known as “perceived persistence.” When a game feature associated with a persistent element is awarded, at least some of the contents of the persistent element are visually removed and the accumulation of value in that persistent element during subsequent game spin cycles resumes from that point. Each of the persistent elements **212**, **214** and **216** exhibits perceived persistence.

[0044] FIG. **6** illustrates the results of a game spin in which accumulation symbols have landed in the array. At step **324**, the game-logic circuitry animates, via the one or more presentation devices, an addition of any coin in the array to the persistent element associated with the coin. To show the transfer of a coin symbol to a persistent element, the coin may be animated to “fly” from the array to land in its associated persistent element. To represent the gradual addition of coins to the persistent element, the persistent element and/or the volume of coins therein may appear to grow in size. In accordance with some embodiments, the persistent element may be an object of fixed size accompanied by some other indication of accumulating value, for example, by a gradual change in color shading, for example, from light red to dark red. In other embodiments, the persistent element may change size and also show some other indication of increasing value, for example, the color of the display in the immediate area of the persistent element may gradually change as its value increases. The size of a persistent element may or may not indicate the likelihood that its associated game feature will be awarded. This will be discussed further below.

[0045] In one or more embodiments, the coin symbols may be carried on the reels as “stickers” applied over an underlying standard reel symbol. As part of the animation sequence at step **324**, in which the coin is transferred to its respective persistent element, the coin “sticker” is removed from its location in the reel array to reveal the standard reel symbol underneath. In other embodiments, the coin symbols may simply be removed from the array and replacement symbols may be randomly selected to take their places. In some embodiments, the coin symbols may remain in the array as “blocking symbols” that break up other potentially winning combinations of standard symbols. These approaches or combinations thereof all fall within the spirit and scope of the invention.

[0046] In FIG. **6**, each transfer of a coin to the persistent elements **212**, **214**, **216** is represented by a “coin” circle and an arrow. See, for example, the arrow from the BLUE coin in the second column of the array **210**, the arrow from the GOLD coin in the fourth column of the array **210**, and the arrow from the RED coin in the fifth column of the array **210** in FIG. **6**. Though not shown in FIG.

6, no coins, coins of only one or two colors or multiple coins of the same color/type may appear in the array at the same time. Animation representing the transfer each coin from the array **210** to its respective persistent element may be presented sequentially or in parallel.

Awarding of Game Features

[0047] At step **326**, the game-logic circuitry randomly determines, via the RNG, whether or not to award the game features associated with any coin symbols detected at step **310**. This random determination is independent of any prior wagering game cycles. A single determination may be made whether to award all of the game features associated with persistent elements that underwent accumulation at step **324** or step **326** may simply test to see whether it is possible that an RNG determination made at each of steps **328**, **33** and **336** may award a game feature.

[0048] In accordance with one or more embodiments, the odds of awarding the game features with a single RNG determination at step **326** may increase according to the number of coins appearing in the array. This may be accomplished, for example, by changing weights associated with the random determination. In other embodiments, the appearance of multiple coins in the array has no effect on the probability of awarding the game features associated with the persistent elements corresponding to the coins appearing in the array. If the game features are awarded, for each coin symbol detected at step **310**, an indication such as a “WINNER!” label may appear next to each respective coin's associated persistent element, or the respective persistent element may be otherwise highlighted. Persistent elements associated with non-awarded game features may be further deemphasized, for example, by shading or “greying” them out. FIG. **7** illustrates an example of a game spin in which only the RED game feature has been won.

[0049] If it is determined at step **326** that no game features will be awarded, the game-logic circuitry returns to FIG. **3**A and proceeds to step **312**, described further below. If, however, at least one game feature will be awarded, the game-logic circuitry instead proceeds to step **328** to begin the process of awarding the one or more game features.

[0050] At step **328**, if at least one BLUE coin associated with the first persistent element **212** was present in the array **210** at step **310** and it was determined at step **326** that the BLUE game feature would be awarded, the game logic circuitry awards the game feature associated with the first persistent element **212** at step **330**. As mentioned above, in some embodiments, step **328** may also include a separate RNG determination of whether or not to award the BLUE game feature. If such an additional determination is made at step **328** and it is determined that the BLUE game feature will not be awarded, the game-logic circuitry continues the flow at step **332** and step **330** is not performed.

[0051] In the examples shown in FIGS. **4-8**, the game feature associated with the first persistent element **212** is “Respin.” This exemplar game feature holds certain symbols in place in the array, for example all K and WILD symbols, and respins the reels to land new symbols in the remaining locations in the array to determine a final outcome for the spin. Thus, the “Respin” game feature is an “instant” game feature that is applied only to the current game outcome. Once the “Respin” game feature has been awarded at step **330**, including modification of the landed symbols in the array, the game logic circuitry proceeds to step **332**.

[0052] At step **332**, if at least one GOLD coin associated with the second persistent element **214** was present in the array **210** at step **310** and it was determined at step **326** that the GOLD game feature would be awarded, the game logic circuitry awards the game feature associated with the second persistent element **214** at step **334**. As mentioned above, in some embodiments, step **332** may also include a separate RNG determination of whether or not to award the GOLD game feature. In these situations, if, at step **332**, it is determined that the GOLD game feature will not be awarded, the game-logic circuitry continues the flow at step **336** and step **334** is not performed.

[0053] In the example shown, the game feature associated with the second persistent element **214** is “Wild Stack,” in which a full stack of symbols on a reel will be replaced by wild symbols and held in place on the next spin. Thus, the “Wild Stack” game feature is a “deferred” game feature that is applied to and affects the next game spin. Once the “Wild Stack” game feature has been awarded at step **334**, the game logic circuitry proceeds to step **336**.

[0054] At step **336**, if at least one RED coin associated with the third persistent element **216** was present in the array **210** at step **310** and it was determined at step **326** that the RED game feature would be awarded, the game logic circuitry awards the game feature associated with the third persistent element **216** at step **338**. Again, in some embodiments, step **336** may also include a separate RNG determination of whether or not to award the RED game feature. In these cases, if, at step **336**, it is determined that the RED game feature will not be awarded, the game-logic circuitry continues the flow at step **340** and step **338** is not performed.

[0055] In the example shown, the game feature associated with the third persistent element **216** is a “Free Spins” game feature. FIG. **7** illustrates an example of a game outcome that awards free spins. In some embodiments, the number of free spins is predetermined. A certain number of free spins, for example, five free spins, will be initiated with the number of free spins remaining tracked by a free spin counter **230**. In other embodiments, the number of free spins may be randomly determined and vary between each awarding of free spins. The free spins may be played in a “hold and spin” manner where certain types of symbols, once landed in the array, are held in place and may variously persist in the array for at least one additional free spin cycle, until they contribute to a winning outcome, until all of the free spins have been played, etc. In some embodiments, wild symbols or other symbols that may improve the chances of winning or provide higher pays may also be held. Like the “Wild Stack” game feature, the “Free Games” feature is a “deferred” game feature. Once applied, as described below with respect to step **314**, reel strip layouts may remain the same as those of the base game or may be different. Once the game feature has been awarded at step **338**, the game logic circuitry proceeds to step **340**.

Applying Non-Deferred Game Features

[0056] Once all of the eligible game features have been awarded at one or more of steps **330**, **334** and **338**, subsequent games will be played according to game rules and conditions set by the one or more awarded game features, enhancing the excitement and EV of the game.

[0057] At step **340**, any non-deferred game features are applied. Any deferred game features will be applied later (at step **316**). In this example, if the “Respin” game feature was awarded at step **330**, the reels will be respun and any locations in the array not held in place will receive newly landed symbols. Thus, the game feature is applied to the original outcome of the current game cycle prior to win evaluation. The game logic circuitry then returns to FIG. **3**A, step **312**.

Win Evaluation

[0058] Once any animation of persistent element growth has occurred, any game features have been awarded and any instant game features applied, the game-logic circuitry evaluates the symbols in the array at step **312**. Payouts are awarded in accordance with pay rules. The pay rules may include a pay table. The pay table may, for example, include “line pays,” “ways pays” and “scatter pays.” Line pays occur when a predetermined type and number of symbols appear along an activated payline, typically in a particular order such as left to right, right to left, top to bottom, bottom to top, etc. Ways pays appear on adjacent reels without the requirement to be on a specified pay line or directly adjacent to one another. Scatter pays occur when a predetermined type and number of symbols appear anywhere in the displayed array without regard to position or paylines. Each payline preferably consists of a single symbol position in each column of the array. The number of paylines may be as few as one or as many as possible given each payline consists of a single symbol position in each column of the array. To animate a standard pay, the display may apply a border, pattern, color change, background change, watermark, or other distinguishing characteristic to the winning payline and/or winning symbols that contributed to the pay. For the sake of simplicity, the examples shown here are line pays. FIG. **8**, for example, depicts a line pay of three K symbols in the bottom row of the array **210**. The awarded pay is added to a win meter **204**.

Applying Deferred Game Features

[0059] At step **314**, a check for any awarded but deferred game features is performed. If no deferred game features have been awarded, flow proceeds to step **318**, otherwise, the deferred game features are applied at step **316**. Because deferred game features are applied to the game after win evaluation, deferred game features apply to one or more subsequent game spins.

[0060] For example, if the "Wild Stack" game feature was awarded at step **334**, the symbols in the array positions containing the triggering stack of symbols will be replaced with wild symbols to be held in place when the next reel spin occurs at step **308**. If current game cycle is a regular paid game cycle financed by a wager (a base game), this deferred game feature will be applied to the next paid game spin.

[0061] In another example, if the "Free Spins" game feature was awarded at step **338**, the free spin counter **230** will be initialized and the game placed in a free spin state. Other free spin rule changes or reel layouts may also be applied at this time and will be in effect for the duration of the free spins. Thus, this game feature also applies to subsequent game spins, the primary difference between this feature and the "Wild Stack" feature being that the subsequent spins do not require a wager.

[0062] It should be noted that multiple game features may be awarded if coins of more than one color appear simultaneously in the array **210**. Thus, one, two or three game features may be awarded as the result of a single spin and complement one another in increasing the excitement of the game. For example, in the case where all three game features are simultaneously awarded, the current array **210** will be modified per the "Respin" feature, any wins will be paid, then the "Wild Stack" modification will be applied to the reels in advance of the next spin, which, because the "Free Spins" game feature was awarded, will also not require placement of a wager.

[0063] At step **318**, the game-logic circuitry checks to see whether the game is in a free spin state as a result of free games having been previously initiated at step **316** in either the current or a prior game cycle. If the game is in a free game state, the game logic circuitry returns to step **308** for another spin without requiring a wager.

Game Play Continuation/Termination

[0064] At step **320**, the game-logic circuitry determines whether or not it has received a cashout input via at least one of the one or more player input devices of the gaming machine. If it has not received a cashout input, the game-logic circuitry waits for the next wager input at step **306**. If it has received a cashout input, the game-logic circuitry initiates a payout from the monetary balance on the meter **200** in FIGS. **4-8**. The data processing method then ends at step **322**.

[0065] The recitations of a value input device for establishing a credit balance, an input device for accepting a wager input that initiates a spin, and a value output device for paying out the credit balance are integrally incorporated within the steps of the data processing method. For example, the presentation of game outcomes through the spinning and stopping of the reels is essential to the game outcome determinations, which may only be initiated by the accepted wager input. Furthermore, a value input device for establishing a credit balance, an input device for accepting a wager that initiates a spin, and a value output device for paying out the credit balance are physical, structural elements that are not shared by generic or well-known computing devices but, rather, are particular to gaming machines.

[0066] Embodiments of the present invention realize benefits in increased computer processing efficiency with minimized processing overhead, fewer rules to be evaluated, fewer player inputs to be monitored, and simpler graphical representations. With respect to the game feature awarding process, if accumulation condition appears in the array at step **310**, the game-logic circuitry foregoes any random determination of whether a game feature will be awarded. Furthermore, if any coin symbols do appear in the array, regardless of the number of pots and associated game features that may be won in parallel, only a single invocation of the RNG at step **326** is required to determine whether any game feature will actually be awarded. In contrast, in typical prior art systems with mystery bonus triggers, the game-logic circuitry makes a random determination in each and every wagering game cycle and for each and every game feature that may be won, thereby reducing processing efficiency compared to the method presented herein.

[0067] In this description, numerous specific details are set forth. However, it is understood that embodiments of the invention may be practiced without these specific or with slightly varying details.

[0068] For example, in accordance with one or more embodiments, the displayed size of a perceived persistence/true persistence persistent element may partially represent the likelihood that its associated game feature will be awarded.

[0069] In another example, in accordance with still other embodiments, each persistent element need not be of the same form, for example, the triangles of FIGS. **4-8** may vary from one persistent element to the next, for example, persistent element may be represented as triangle, one as a square and one as a circle. As a further example, FIG. **9** illustrates that two of the persistent elements **212**, **216** may be portrayed as unicorns, while one persistent element **214** may be portrayed as a castle.

[0070] In accordance with some embodiments, certain combinations of symbols may take the place of the discrete designated symbols such as the BLUE, GOLD and RED accumulation symbols of FIGS. **4-8**. In one example, a winning combination of designated symbols, in addition to earning an award, may increase the perceived persistence aspect of an associated persistent element and potentially trigger the awarding of its associated game feature. To illustrate, a winning combination of a designated symbols, as represented by a box around three unicorn symbols in the bottom row of FIG. **10**, increases the size of the first persistent element **212** and potentially triggers awarding the “Respin” game element. In the example of FIG. **10**, the first persistent element **212** has increased in size compared to its size in FIG. **9** while the other two persistent elements **214** and **216** have remained the same size. In addition, the “Respin” game element has been awarded, as indicated by the highlighted nature of its label in FIG. **10** compared to its appearance in FIG. **9**. In the example of FIG. **10**, as a non-deferred game feature, the application of the “Respin” game feature has held all unicorn and wild symbols (the “pearl” symbol in the fourth column of the array) and the reels are currently being re-spun as described above with respect to step **330**.

[0071] In another example, the combination of symbols need not be a winning combination of symbols. In the illustration of FIG. **11**, a full stack of unicorn symbols appears in the first column of the array. A full column of a designated symbol may be defined as the triggering combination for increasing the size of a persistent element, in this case, the third persistent element **216**, and potentially awarding its associated game feature. In the example of FIG. **11**, the third persistent element **216** has increased in size compared to its size in FIG. **9**. In addition, the deferred “Wild Stack” game element has been awarded, as indicated by the highlighted nature of its label in FIG. **11** compared to its appearance in FIG. **9**. In the example of FIG. **12**, which represents the next paid spin, application of the “Wild Stack” game feature has held the triggering stack of symbols in place and replaced them with wild symbols, as indicated by the stack of “pearl” symbols in the first column of the array. The reels are currently being re-spun as described above with respect to step **334**.

[0072] In other instances, well-known circuits, structures and techniques have not been shown in detail in order not to obscure the understanding of this description. Note that in this description, references to “one embodiment” or “an embodiment” mean that the feature being referred to is included in at least one embodiment of the invention. Further, separate references to “one embodiment” in this description do not necessarily refer to the same embodiment; however, neither are such embodiments mutually exclusive, unless so stated and except as will be readily apparent to those of ordinary skill in the art. Thus, the present invention can include any variety of combinations and/or integrations of the embodiments described herein. Each claim, as may be amended, constitutes an embodiment of the invention, incorporated by reference into the detailed description. Moreover, in this description, the phrase “exemplary embodiment” means that the embodiment being referred to serves as an example or illustration.

[0073] Block diagrams illustrate exemplary embodiments of the invention. Flow diagrams illustrate operations of the exemplary embodiments of the invention. The operations of the flow diagrams are described with reference to the example embodiments shown in the block diagrams. However, it should be understood that the operations of the flow diagrams could be performed by embodiments of the invention other than those discussed with reference to the block diagrams, and embodiments discussed with references to the block diagrams could perform operations different than those discussed with reference to the flow diagrams. Additionally, some embodiments may not perform all the operations shown in a flow diagram. Moreover, it should be understood that although the flow diagrams depict serial operations, certain embodiments could perform certain of those operations in parallel or in a different sequence.
[0074] Each of these embodiments and obvious variations thereof is contemplated as falling within the spirit and scope of the claimed invention, which is set forth in the following claims. Moreover, the present concepts expressly include any and all combinations and subcombinations of the preceding elements and aspects.

Claims

1. A gaming machine comprising: a presentation assembly configured to display an array of symbol-bearing reels and at least one persistent element, each persistent element associated with a respective accumulation condition and a distinct game feature; game-logic circuitry configured to: determine satisfaction of accumulation conditions during gameplay and animate accumulation of each persistent element upon satisfaction of its associated accumulation condition; randomly determine awarding of the distinct game feature associated with each persistent element upon satisfaction of the respective accumulation condition; initiate a free-spin feature upon awarding a persistent element specifically associated with awarding free spins; and conditionally enhance the initiated free-spin feature by applying modifications to symbol outcomes during the free spins responsive to simultaneous awarding of the free-spin persistent element and at least one additional persistent element associated with modifying symbol outcomes.
2. The gaming machine of claim 1, wherein the accumulation condition comprises the presence of a single designated reel symbol in a displayed array.
3. The gaming machine of claim 1, wherein the accumulation condition comprises a winning combination of designated symbols.
4. The gaming machine of claim 1, wherein the accumulation condition comprises a non-winning combination of designated symbols.
5. The gaming machine of claim 4, wherein the non-winning combination of designated symbols comprises a stack of identical symbols.
6. The gaming machine of claim 1, wherein the enhancement applied during free spins modifies symbol outcomes by substituting one or more displayed symbols with alternative symbols.
7. The gaming machine of claim 1, wherein the enhancement applied during free spins modifies symbol outcomes by awarding multipliers.
8. The gaming machine of claim 1, wherein the awarded number of free spins is randomly determined.
9. The gaming machine of claim 1, wherein the game-logic circuitry resets the awarded persistent element to an initial state following the awarding of its associated game feature.
10. The gaming machine of claim 1, wherein animations associated with each persistent element include audio effects synchronized with visual animations.
11. A method of operating a gaming machine, the gaming machine comprising a presentation assembly, a value input device, a value output device, and game-logic circuitry, the method comprising: accepting, via the value input device, a wager to initiate gameplay; displaying, via the presentation assembly, symbol-bearing reels and at least one persistent element, wherein each

persistent element has an associated accumulation condition and a distinct associated game feature; determining, via the game-logic circuitry, satisfaction of accumulation conditions during gameplay, animating persistent element accumulation responsive thereto, and randomly awarding associated game features; initiating, via the game-logic circuitry, a free-spin feature upon awarding a persistent element associated with awarding free spins; and conditionally enhancing the free-spin feature by modifying symbol outcomes during the awarded free spins upon simultaneous awarding of at least one additional game feature associated with modifying symbol outcomes.

12. The method of claim 11, wherein the accumulation condition comprises the presence of a single designated reel symbol in a displayed array.

13. The method of claim 11, wherein the accumulation condition comprises a winning combination of designated symbols.

14. The method of claim 11, wherein the accumulation condition comprises a non-winning combination of designated symbols.

15. The method of claim 14, wherein the non-winning combination of designated symbols comprises a stack of identical symbols.

16. The method of claim 11, wherein the enhancement applied during free spins modifies symbol outcomes by substituting one or more displayed symbols with alternative symbols.

17. The method of claim 11, wherein the enhancement applied during free spins modifies symbol outcomes by awarding multipliers.

18. The method of claim 11, wherein the awarded number of free spins is randomly determined.

19. The method of claim 11, further comprising resetting the awarded persistent element to an initial state following the awarding of its associated game feature.

20. The method of claim 11, wherein animations associated with each persistent element include audio effects synchronized with visual animations.
